

Warsaw University of Technology

Faculty of Mathematics
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Project Documentation

A system for reconstructing fragmented documents based on the shape and content of their parts

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Change History

Version	Date	Author	Description of Change
1.0	13.10.2026	ZZ	Baseline version for supervisor sign-off, problem formulation and project scope
1.1	14.10.2026	ZZ	Added functional and non-functional requirements
1.2	15.10.2026	MS	Added risk register and contingency plans
1.3	16.10.2026	HS	Expanded out-of-scope and glossary sections
1.4	18.10.2026	ZZ	Created schedule
1.5	19.10.2026	HS	Assigned roles to RACI matrix
1.6	20.10.2026	MS	Added schedule and RACI matrix to the document
1.7	20.10.2026	HS	Expanded research problem and scope sections
1.8	20.10.2026	ZZ	Added Gantt diagram
1.9	20.10.2026	HS	Added title page and updated text formatting

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Abstract

This document defines the objectives, scope, requirements, and planned execution of the project. It establishes the engineering problem and system vision, specifies the minimum viable product, defines functional and non-functional requirements, and describes the project schedule, team responsibilities, and risk management strategies. It also provides a glossary of key terms. The document serves as a shared reference for the project team and supervisor, supporting consistent implementation, progress monitoring, and verification that the developed system meets the agreed objectives.

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General Description of the System

Context and Engineering Problem

Fragmented documents are often difficult to reconstruct: their pieces are stripped of the context provided by the original page layout, and their number makes the task challenging to perform manually. Reconstructing the original document requires first identifying individual fragments on the source image, and then determining how they fit together.

This project addresses the engineering problem of designing a system that can reconstruct fragmented document images using information derived from the shape and content of their pieces. The system will also generate fragmented document images, allowing reconstruction algorithms to be tested on controlled inputs.

Vision of the System

The vision is to provide a local application that supports the complete workflow of document fragmentation and reconstruction. Users will be able to split document images into pieces, preview the generated fragments, reconstruct documents from fragmented images using a variety of algorithms, and save the resulting images.

The system is intended to support experimentation with different fragmentation and reconstruction algorithms while keeping the workflow accessible through a consistent user interface.

Business Goal

The business goal is to deliver a usable, self-contained tool that reduces the manual effort required to prepare fragmented document images and evaluate document reconstruction methods. By combining fragment generation, reconstruction, and result inspection in one application, the system will support the assessment of alternative algorithms.

Research Goals

The research goals are to:

- investigate shape-based and content-based approaches to document reconstruction;
- compare the performance and reconstruction quality of selected algorithms under different input conditions;
- investigate how fragment rotation, fragment count, and the types or shapes of the fragments affect reconstruction results;
- identify suitable evaluation metrics and heuristics for selecting reconstruction algorithms.

Engineering Project Goal

The objective of this project is to design, implement, and evaluate a system capable of reconstructing fragmented (shredded or torn) documents based on an image of their pieces, as well as artificially splitting a document into such pieces.

Success Criteria

The project will be considered successful when:

- both the document splitting and document reconstruction modules are implemented within the agreed MVP scope;
- the system supports single-image and batch processing where specified in the functional requirements;
- the quantifiable non-functional requirements are tested and the results are documented;
- the reconstruction algorithms are evaluated using documented test cases and appropriate quality measures;
- the application can operate offline, with image processing performed locally.

Project Scope

Minimum Viable Product

1. **Document Splitting Module (Module 1):**
   - Supports single and batch mode processing.
   - Allows users to split document images using various algorithms, preview piece borders, re-generate if needed, and apply random rotations and non-overlapping background placement.
   - Saves the final image of generated fragments locally.
2. **Document Reconstruction Module (Module 2):**
   - Supports single and batch mode processing.
   - Extracts individual fragments from an input image.
   - Reconstructs original documents using a variety of algorithms.
   - Displays the reconstructed output, allows for iterative improvement (adjusting either the processing time or algorithm used), and saves the final result locally.

Out of Scope (Future Work)

1. Integration with external cloud storage providers or dedicated mobile applications.
2. Reconstruction of documents with missing fragments.
3. Reconstruction of several documents whose fragments are combined.

Functional Requirements

US-01: Loading a document image for splitting

- **As a:** User of the document splitting module,
- **I want to:** Upload a single document image or select a folder of document images,
- **So that:** The system can process them for fragment generation.

Acceptance Criteria (Gherkin):

- **Given:** The splitting module is open,
- **When:** The user uploads a valid image file or selects a folder containing valid image files,
- **Then:** The system loads the images and displays a preview of the first document.
- **Scenario (Invalid input):**
- **When:** The user uploads a file in an unsupported format or a file that doesn't contain any valid image files,
- **Then:** The system displays an error message indicating the accepted formats.

US-02: Splitting a document into fragments

- **As a:** User of the document splitting module,
- **I want to:** Select a splitting algorithm and its parameters and preview the proposed piece borders overlaid on the document,
- **So that:** I can control how the document is fragmented.

Acceptance Criteria (Gherkin):

- **Given:** A document image has been loaded,
- **When:** The user selects a splitting algorithm and confirms its parameters,
- **Then:** The system displays the proposed piece borders overlaid on the original image.
- **When:** The user is not satisfied with the result and clicks "Generate again",
- **Then:** The system generates a new set of borders using the currently selected algorithm and its parameters.

US-03: Generating a fragmented image

- **As a:** User of the document splitting module,
- **I want to:** Confirm the generated borders and have the system randomly rotate the pieces and arrange them on a plain background without overlapping,

- **So that:** I get an image with document fragments which can later be used for testing the document reconstruction module.

  Acceptance Criteria (Gherkin):

  - **Given:** The user has confirmed a set of piece borders,
  - **When:** The user sets the parameters for arranging the pieces and confirms them,
  - **Then:** The system produces an image with all fragments placed on the background without any overlap.
  - **When:** The user selects the option to save the final image and selects a disk location,
  - **Then:** The result is saved to the chosen disk location.

US-04: Extracting fragments from an input image

- **As a:** User of the document reconstruction module,
- **I want to:** Upload an image containing document fragments (single or batch mode),
- **So that:** The system extracts individual pieces for reconstruction.

  Acceptance Criteria (Gherkin):

  - **Given:** The reconstruction module is open,
  - **When:** The user uploads a valid image containing non-overlapping fragments,
  - **Then:** The system identifies and extracts all individual fragments from the image.
  - **Scenario (Invalid input):**
  - **When:** The image does not contain identifiable fragments,
  - **Then:** The system displays an appropriate error message.

US-05: Reconstructing a document from fragments

- **As a:** User of the document reconstruction module,
- **I want to:** Select a reconstruction algorithm (or let the system choose one automatically) and view the reconstructed document,
- **So that:** I can receive the original document content in a legible form.

  Acceptance Criteria (Gherkin):

  - **Given:** Fragments have been extracted from the input image,
  - **When:** The user selects an algorithm or chooses automatic selection, and starts reconstruction,
  - **Then:** The system displays the reconstructed document image.

US-06: Iterative enhancement of reconstruction

- **As a:** User of the document reconstruction module,
- **I want to:** Review the reconstruction result and choose to accept it, continue the same algorithm for more time or switch to a different algorithm,
- **So that:** I can iteratively improve the reconstruction quality.

Acceptance Criteria (Gherkin):

- **Given:** A reconstruction result is displayed,
- **When:** The user selects "Continue" with the current algorithm,
- **Then:** The system resumes processing.
- **When:** The user selects "Change algorithm" and picks a new one,
- **Then:** The system restarts reconstruction with the new algorithm.
- **When:** The user selects "Accept",
- **Then:** The system allows saving the result locally.

Non-Functional Requirements

ID	Category	Target Quantifiable Metric	Verification Method
NFR-01	**Performance (Splitting)**	Splitting a single A4-sized document image into at least 20 fragments completes under 60 seconds.	Automated benchmark script with sample dataset
NFR-02	**Performance (Reconstruction)**	Reconstruction of a document with at least 20 fragments completes under 60 minutes.	Automated benchmark script with sample dataset
NFR-03	**Scalability**	The system handles documents split into up to 100 fragments without crashing.	Stress tess
NFR-04	**Usability**	A user with no prior training can complete a full split-and-reconstruct workflow.	Usability test
NFR-05	**Error Handling**	100% of invalid inputs produce a user-friendly error message.	Test covering all invalid input scenarios

ID	Category	Target Quantifiable Metric	Verification Method
NFR-06	**Availability**	The system operates fully offline	Disconnected network test
NFR-07	**Security**	All image processing is performed locally and no data is transmitted to external servers.	Manual verification of network traffic

Preliminary Risk Register

Project and Technical Risk Register

ID	Risk Event	Likelihood (L/M/H)	Impact (L/M/H/Crit)	Owner	Mitigation Strategy
RSK-01	**Team member illness / drop-out**	L	H	Everyone	Peer code review for every PR, pair programming on the critical path, modular architecture.
RSK-02	**Insufficient accuracy for the reconstructing algorithm**	M	H	Everyone	Explore different algorithms; if the problem persists focus on an analysis of algorithmic limitations and failure cases.
RSK-03	**Environment incompatibility in the MiNI laboratory**	M	Crit	MS	Test deployment on lab machines after implementing Module 1 and another after Module 2, not the night before acceptance.
RSK-04	**Hardware failure (loss of data / code on laptop)**	M	Crit	Everyone	Pushes to remote repository (GitHub) after each change.
RSK-05	**Thesis writing workload accumulation in January**	H	H	Everyone	Incremental drafting of thesis chapters; peer reviews within the team.

ID	Risk Event	Likelihood (L/M/H)	Impact (L/M/H/Crit)	Owner	Mitigation Strategy
RSK-06	**Unacceptably long execution time**	H	M	HS	Implementing early stopping to the algorithms and accepting the worse results.
RSK-07	**Rotation of the fragments resulting in quality loss and document illegibility.**	M	M	ZZ	Remembering the fragments' original visual, rotating it only once to match the found rotation and accepting the loss of information resulting in this operation.
RSK-08	**Loss of information on the edges of fragments as a result of inacurate cut or assembling.**	M	L	ZZ	Accepting the problem as an opportunity to test the algorithms in a real-world scenario (manual paper shredding).
RSK-09	**Excessive RAM requirements for running the program smoothly**	M	M	MS	Switching to the faculty MiNI GPU cluster
RSK-10	**Excessive hyperparameter tuning time**	M	M	HS	Finding a different algorithm or accepting the suboptimal hyperparameters.
RSK-11	**No measurable metrics for assessing the accuracy of splitting and reconstructing the document.**	H	M	ZZ	Relying solely on manual visual assessment.
RSK-12	**Inability of the chosen UI library to meet our design requirements**	L	L	MS	UI redesign or library change.
RSK-13	**To high difficulty of the chosen UI library**	L	L	MS	Library change.

Contingency Plans for High-Risk Modules

Contingency Protocol 1: Pre-recorded Demo Failure during Defense

- **Plan A (Primary):** Run a pre-recorded video demo with live narration during the 10-minute presentation.
- **Plan B Trigger:** Sound system issues, network failure.
- **Plan B (Contingency):** Immediately switch to a presentation with screenshots showcasing the application's core functionality.

Contingency Protocol 2: Literature Dataset Issues

- **Plan A (Primary):** Use existing datasets from literature.
- **Plan B Trigger:** Restricted access to external datasets or unsatisfactory results from them.
- **Plan B (Contingency):** Construct a custom dataset by shredding and scanning physical documents.

Schedule and Role Allocation

Schedule

WBS	Task	Duration (Days)	Start Date	End Date
1	**Project Planning & Architecture**	**13**	**11-10**	**23-10**
1.1	Define business objective	3	11-10	13-10
1.2	Define system vision	3	11-10	13-10
1.3	Define functional requirements	4	13-10	16-10
1.4	Define non-functional requirements	3	14-10	16-10
1.5	Perform risk analysis	2	14-10	15-10
1.6	Prepare project schedule and divide tasks between team members	4	16-10	19-10
1.7	Decide on the formats in which data will be stored and handled	1	16-10	16-10
1.8	Define system architecture and main classes	5	19-10	23-10
1.9	Decide on main dependencies (e.g. external libraries)	8	13-10	20-10
1.10	Prepare a shortlist of algorithms to be implemented for each module	11	13-10	23-10
2	**Module 1: Document Fragmentation**	**13**	**19-10**	**31-10**
2.1	Implement input handling	2	19-10	20-10

WBS	Task	Duration (Days)	Start Date	End Date
2.2	Implement fragmentation algorithms	10	19-10	28-10
2.3	Implement the random rotation of each fragment	3	21-10	23-10
2.4	Implement fragment placement on background	3	21-10	23-10
2.5	Implement image export	2	23-10	24-10
2.6	Write unit tests for module 1	3	26-10	28-10
2.7	Validate results through subjective observation	9	23-10	31-10
3	**Module 2A: Fragment Extraction**	**7**	**1-11**	**7-11**
3.1	Implement the algorithm for finding fragment borders	4	1-11	4-11
3.2	Implement extracting the fragments into separate objects	2	4-11	5-11
3.3	Write unit tests for module 2A	2	6-11	7-11
4	**Module 2B: Document Reconstruction**	**17**	**8-11**	**24-11**
4.1	Implement shape-based reconstruction algorithms	13	8-11	20-11
4.2	Implement content-based reconstruction algorithms	13	8-11	20-11
4.3	Write unite tests for module 2B	4	21-11	24-11
5	**Benchmarking**	**6**	**25-11**	**30-11**
5.1	Implement the extraction of text from a reconstructed document	3	25-11	27-11
5.2	Prepare benchmarking scenarios for various input types	3	25-11	27-11
5.3	Perform automated benchmarking of reconstruction algorithms using automated text extraction	3	27-11	29-11
5.4	Perform manual benchmarking of reconstruction algorithms based on final document readability	4	26-11	29-11
5.5	Construct heuristics for algorithm selection	2	29-11	30-11
6	**Integration, UI**	**34**	**1-11**	**4-12**
6.1	Integrate modules into an unified pipeline	4	1-12	4-12
6.2	Building UI	30	1-11	30-11
7	**Documentation & Thesis writing**	**62**	**1-12**	**31-01**
7.1	Compile existing documentation and benchmarking results into initial thesis draft	20	1-12	20-12
7.2	Write core chapters for thesis	38	1-12	7-01
7.3	Final buffer time, thesis polishing and defense presentation	25	7-01	31-01

Gantt diagram

Figure 1 represents the Gantt Diagram corresponding to the schedule.

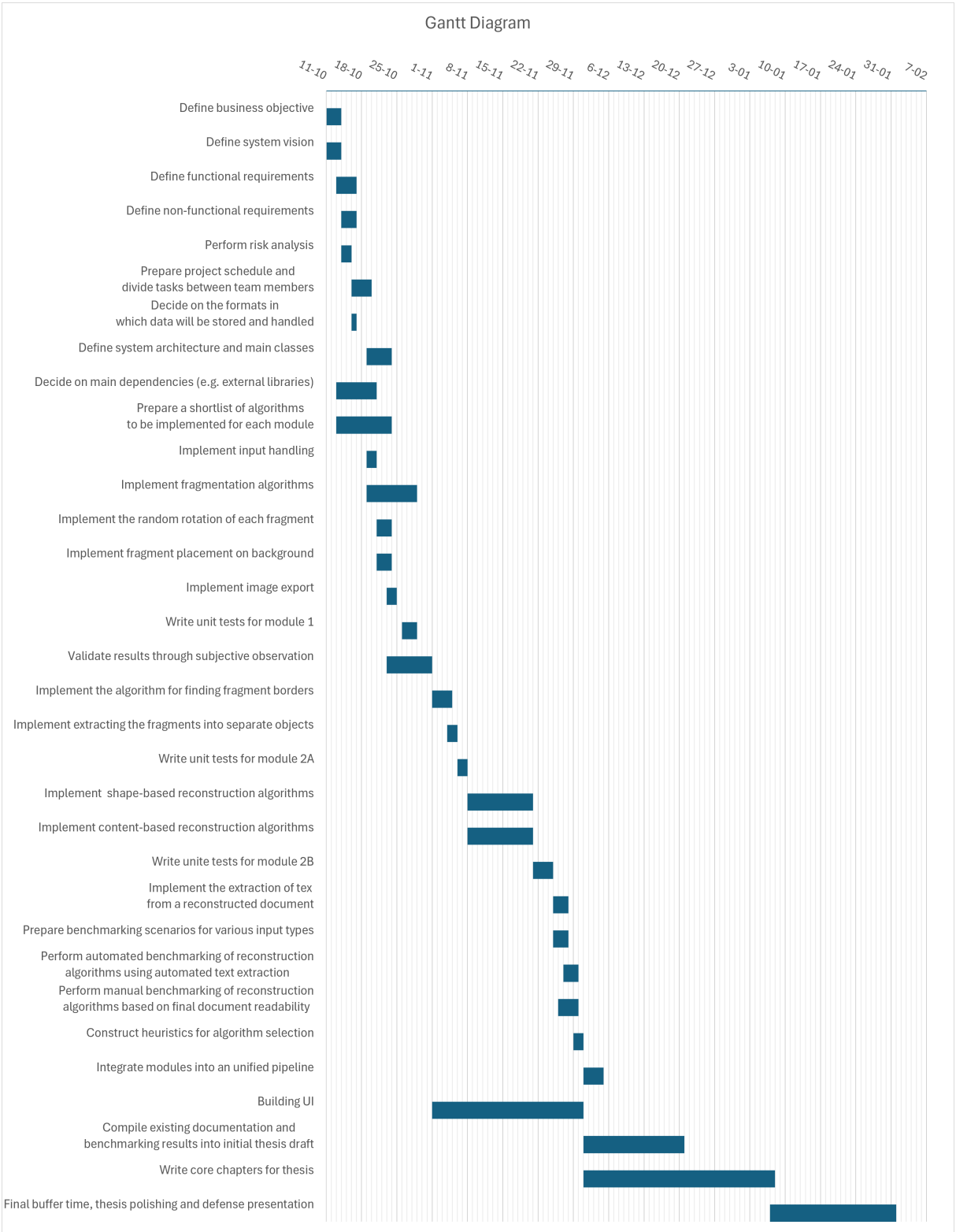


Figure 1: Project Gantt Diagram

RACI Matrix

WBS	Responsible	Accountable	Consulted	Informed
1				
1.1	MS, HS, ZZ	HS	JR	—
1.2	MS, HS, ZZ	HS	—	JR
1.3	MS, HS	HS	ZZ	JR
1.4	ZZ	ZZ	MS, HS	JR
1.5	MS, ZZ	ZZ	HS	JR
1.6	HS, ZZ	ZZ	MS	—
1.7	MS, HS	MS	ZZ	—
1.8	MS, HS, ZZ	MS	—	—
1.9	MS	MS	HS, ZZ	—
1.10	MS, HS, ZZ	MS	JR	—
2				
2.1	ZZ	ZZ	MS, HS	—
2.2	MS, HS, ZZ	HS	—	—
2.3	MS	MS	HS, ZZ	—
2.4	HS	HS	MS, ZZ	—
2.5	ZZ	ZZ	MS, HS	—
2.6	MS	MS	HS, ZZ	—
2.7	HS	HS	MS, ZZ	JR
3				
3.1	HS	HS	MS, ZZ	—
3.2	MS	MS	HS, ZZ	—
3.3	ZZ	ZZ	MS, HS	—
4				
4.1	MS, HS, ZZ	HS	—	JR
4.2	MS, HS, ZZ	MS	—	JR
4.3	ZZ	ZZ	MS, HS	—
5				
5.1	ZZ	ZZ	MS, HS	—
5.2	HS	HS	MS, ZZ	—
5.3	MS	MS	HS, ZZ	—
5.4	MS, HS, ZZ	ZZ	—	—
5.5	HS	HS	MS, ZZ	JR
6				
6.1	ZZ	ZZ	MS, HS	JR

WBS	Responsible	Accountable	Consulted	Informed
6.2	MS, HS, ZZ	MS	HS, ZZ	JR
7				
7.1	MS, HS, ZZ	ZZ	—	—
7.2	MS, HS, ZZ	HS	JR	—
7.3	MS, HS, ZZ	MS	—	JR

Glossary

Term	Definition
APD (Archiwum Prac Dyplomowych)	University Thesis Archive; official web portal of Warsaw University of Technology used for thesis submission, anti-plagiarism verification, and archiving.
MVP (Minimum Viable Product)	The minimal functional scope of the project fulfilling all foundational requirements for academic project evaluation and defense.
Module 1	The document-splitting module, whose scope includes splitting an image into several fragments and arranging those fragments against a background.
Module 2	The document reconstruction module, whose scope includes extracting individual fragments from an image and reconstructing the original document based on the fragments' shape and content.